MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)

STANDARD OPERATING PROCEDURE: MRPL-SOP-VDU-204 (Rev 3.0)

Title: Vacuum Distillation Unit (VDU-2) Column Startup, Ejector Vacuum Pull & Flare Safety Interlocks

Classification: Confidential | **Target Unit:** VDU-2 & Utilities | **Effective Date:** 2026-09-01

1. Purpose & Scope

This procedure establishes the step-by-step operating guidelines for pre-commissioning nitrogen leak testing, vacuum pull sequence via three-stage steam ejector systems, and furnace transfer line heating under vacuum column operation. All steps adhere to ASME Section VIII Div 1 rules and ISA-5.1 safety protocols.

2. Critical Operating Parameters & Safety Trip Limits

Parameter / ISA Tag	Nominal Range	High Alarm Limit	Safety Action / Interlock
Column Top Vacuum (PIC-209)	12 – 15 mmHg abs	> 25 mmHg abs	Trip Residue Feed, Open N2 Bleed
Furnace Transfer Temp (TIC-205)	385°C – 395°C	> 410°C	Fuel Gas Trip Valve XV-202 Closes
Steam Ejector Motive (PRV-802)	10.5 – 12.0 bar	< 9.0 bar	Switch to Standby Ejector Jet E-202B
Column Bottoms Liquid (LIC-201)	45% – 60%	> 75%	Auto-increase HVGO product pump speed
NDT Wall Thickness (Ultrasonic)	36.4 mm nominal	< 32.5 mm	Turnaround overhaul required

3. Standard Operating Sequence

Step 3.1 (Nitrogen Leak Test): Pressurize the vacuum column to 0.8 bar(g) with high-purity nitrogen via valve FCV-201. Verify soap bubble tightness on all manway gaskets M-201 through M-208. Hold pressure for 60 minutes with < 0.05 bar drop.

Step 3.2 (Steam Ejector Vacuum Pull): Open motive steam isolation valve at 11.5 bar to first-stage booster ejector E-201A. Monitor vacuum gauge PIC-209 until column pressure decreases below 20 mmHg abs within 45 minutes.

Step 3.3 (Atmospheric Residue Feed Admission): Start feed booster pump P-204A at 180 m3/h when transfer line temperature TIC-205 reaches 340°C. Gradually stabilize heavy vacuum gas oil (HVGO) draw-off rate at 65 m3/h.

4. Emergency Shutdown & Depressurization

In the event of vacuum loss (> 35 mmHg abs) or furnace tube rupture, immediately depress the manual ESD button at DCS console 04. This trips solenoid valve SOV-201, isolating fuel gas within 1.2 seconds and opening emergency nitrogen purge FCV-205.